Marmara Lectures

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Dirac delta function

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Introduction to the dirac delta function

A dirac delta $\delta(t)$ is a function which

- has area one on top of every interval which contains the point t=0, ie, $\int_a^b \delta(t)dt = 1$ if $0 \in [a, b]$
- has area zero on top of every interval which doesnt contain the point t=0, ie, $\int_c^d \delta(t)dt = 0$ if $0 \notin [c, d]$

Let us open this definition, and investigate its salient points.

Dirac delta puts area 1 on top of the point t=0

We have stated that dirac delta integrates to one for every interval that contains the point $t = 0$. This naturally includes infinitesimal intervals which contains $t = 0$. Hence on interval $[-\varepsilon, \varepsilon]$, we can write

$$\lim_{\varepsilon \to 0} \int_{-\varepsilon}^{\varepsilon} \delta(t)dt = 1$$

As $\varepsilon \to 0$ nearly all points neighbouring $t = 0$ are excluded, but the integral still remains 1. Hence it can be said that all the area is placed on top of the point $t = 0$. Note that this contrasts sharply with the usual piecewise continuous functions that we are accustomed to. Take a function $f(t)$ which has well defined value at the point $t = 0$, $f(0) = f_0$ and which is continuous there. Then

$$\lim_{\varepsilon \to 0} \int_{-\varepsilon}^{\varepsilon} f(t)dt \approx \lim_{\varepsilon \to 0} 2\varepsilon f_0 = 0$$

We can concludre that $\delta(t)$ behaves differently from a normal function around $t = 0$.

Dirac delta is not zero for points $t \neq 0$

Our definition says that dirac delta integrates to zero on any interval $[a, b]$ which does not contain $t = 0$. Or,

$$\int_a^b \delta(t)dt = 0$$

This may give rise to the belief that $\delta(t)$ is zero at everywhere except at $t = 0$. This belief is wrong, and we will give some counterexamples below. But, as a starter, just consider the case

$$f(t) = \lim_{\omega \to \infty} e^{-|t|} \sin(\omega t) U(t)$$

$f(t)$ is definitely not zero, but it will integrate to zero in any given interval by the Riemann-Lebesgue lemma.

Dirac delta is not a function

Though we have called dirac delta "a function" at the start of this chapter, in reality it is not a function. A function is a rule which maps a given value to an another value. If $g(t)$ is a function then $g(2)$ has a well-defined value, say $g(2) = 7$.
Consider $f(t)$ given above. If we try to find $f(2)$

$$f(2) = \lim_{\omega \to \infty} e^{-|2|} \sin(\omega 2)$$

we will see that no such value exists. Such mathematical objects are known as distributions. They are more general than functions. All functions are distributions, but not all distributions are functions. Dirac delta is a distribution. It is not a function (though we will continue to call it a function to keep things simple).

Constructing a Dirac delta functon

Up untill now, we gave a definition of the dirac delta function but did not provide an example. Now, we will construct a simple one. Then we will give a general procedure to construct dirac deltas, and costruct more complicated examples with the help of this general procedure.

Constructing a Dirac delta function: A simple example

Consider the function

$$\frac{1}{2\varepsilon}\chi_{[-1,1]}(\frac{t}{\varepsilon})$$

which can also be written as

$$\frac{1}{2\varepsilon}\chi_{[-\varepsilon,\varepsilon]}(t)$$

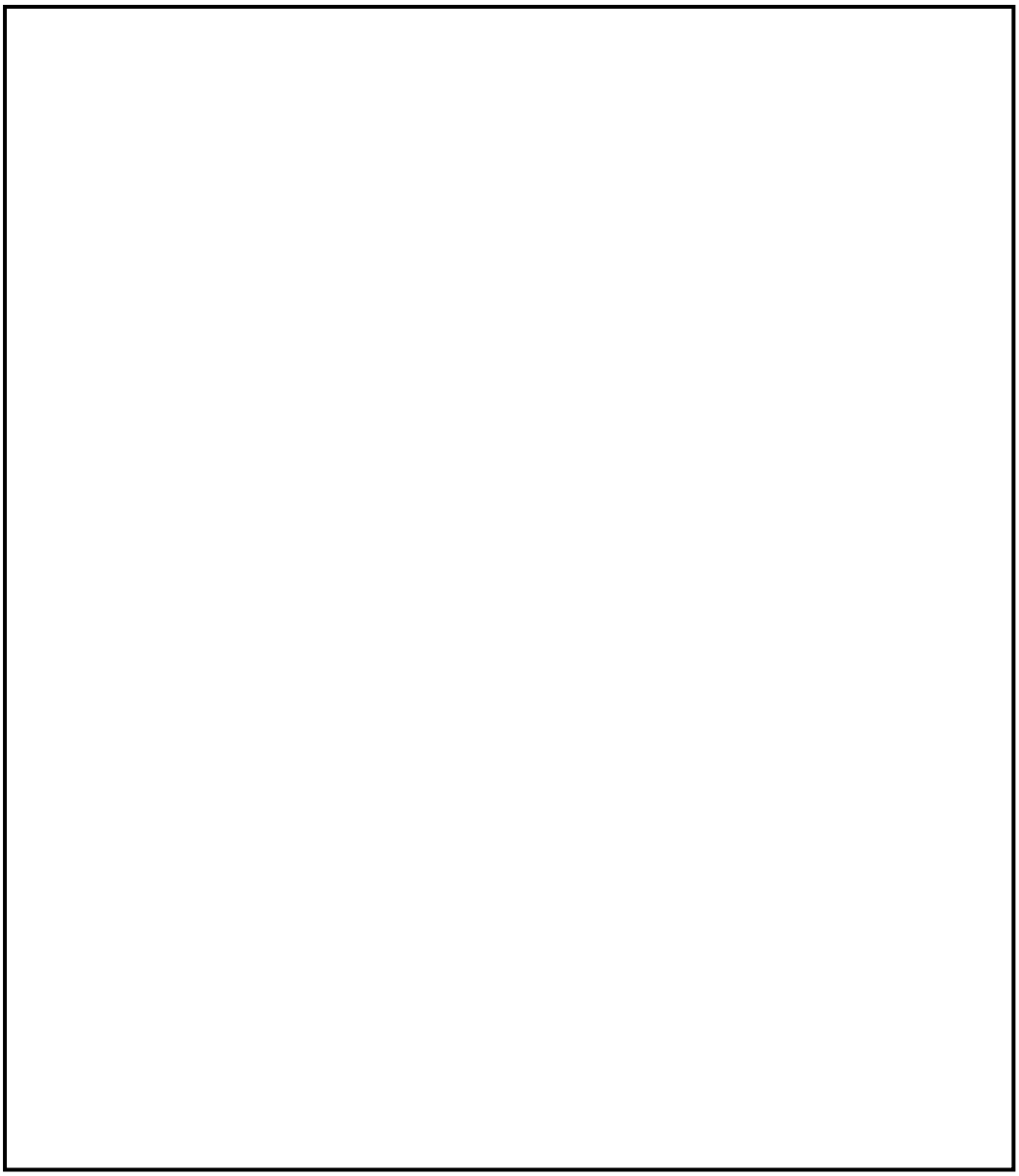
Two properties of this function are

1. The area under it is 1 independent of the value of ε, ie,

$$\int_{-\infty}^{\infty}\frac{1}{2\varepsilon}\chi_{[-\varepsilon,\varepsilon]}(t)dt=\frac{1}{2\varepsilon}\int_{-\varepsilon}^{\varepsilon}1dt=\frac{1}{2\varepsilon}2\varepsilon=1$$

2. All the area of the function is located at the top of the interval $[-\varepsilon,\varepsilon]$. As ε decreases, this area becomes more and more concentrated around the origin. . A plot illustrating this process is given below:

Plot of χ(t)/t



Frequency: 1/

We can conclude that

$$\delta(t) = \lim_{\varepsilon \to 0} \frac{1}{2\varepsilon} \chi_{[-1,1]}(\frac{t}{\varepsilon})$$

Note that the limit is absolutely essential in the above equation. In other words

$$\delta(t) \neq \frac{1}{2\varepsilon} \chi_{[-1,1]} \left(\frac{t}{\varepsilon} \right)$$

even for $\varepsilon = 10^{-1000000000}$. This will be a good approximation, but not the dirac delta itself.

A pitfall is to think that a dirac delta is a function with 0 width and infinite height. This is not correct. Both of the limits below are of $0.\infty$ form:

$$\lim_{\varepsilon\to 0}\frac{1}{2\varepsilon^2}\chi_{[-1,1]}(\frac{t}{\varepsilon})=0.\infty=\infty$$
$$\lim_{\varepsilon\to 0}\frac{1}{2\sqrt{\varepsilon}}\chi_{[-1,1]}(\frac{t}{\varepsilon})=0.\infty=0$$

while we require $0.\infty=1$. So none of the limits above describe a dirac delta function.

Constructing a Dirac delta function: The general Procedure

After considering the simple case discussed above, we can present a method by which a dirac delta can be constructed from any function with finite area.

Let us consider a function $f(t)$ with a finite nonzero area A, ie,

$$\int_{-\infty}^{\infty}f(t)dt=A$$

Then we can claim that

$$\delta(t)=\lim_{\varepsilon\to 0}\frac{1}{A\varepsilon}f(\frac{t}{\varepsilon})$$

Or, equivalently

$$\delta(t)=\lim_{\varepsilon\to \infty}\frac{\varepsilon}{A}f(\varepsilon t)$$

Note that it is not possible to construct dirac delta from functions whose area A is zero.

We will not give a formal proof of this formula, but we will explain the intuition behind it:

1. The area under the construction is 1 for every ε.

$$\int_{-\infty}^{\infty}\frac{1}{A\varepsilon}f(\frac{t}{\varepsilon})dt=\frac{1}{A}\int_{-\infty}^{\infty}f(u)du=\frac{1}{A}A=1$$

Here we have made the change of variables $u = \frac{t}{\varepsilon}$, $du = \frac{dt}{\varepsilon}$.

2. Let the interval $[-\Delta_1, \Delta_2]$, for any $\Delta_1 > 0, \Delta_2 > 0$ contain the point $t = 0$. Then, by using the same change of variables

$$\lim_{\varepsilon \to 0} \int_{-\Delta_1}^{\Delta_2} \frac{1}{A\varepsilon} f(\frac{t}{\varepsilon}) dt = \frac{1}{A} \lim_{\varepsilon \to 0} \int_{-\frac{\Delta_1}{\varepsilon}}^{\frac{\Delta_2}{\varepsilon}} f(u) du = \frac{1}{A} \int_{-\infty}^{\infty} f(u) du = \frac{1}{A} A = 1$$

3. Consider the interval $[\Delta_3, \Delta_4]$, for any $0 < \Delta_3 < \Delta_4$ which does not contain the point $t = 0$. Then the interval $[-\Delta_3, \Delta_4]$ will contain the point $t = 0$ and we can write

$$\lim_{\varepsilon \to 0} \int_{-\Delta_3}^{\Delta_4} \frac{1}{A\varepsilon} f(\frac{t}{\varepsilon}) dt = \lim_{\varepsilon \to 0} \int_{-\Delta_3}^{\Delta_3} \frac{1}{A\varepsilon} f(\frac{t}{\varepsilon}) dt + \lim_{\varepsilon \to 0} \int_{\Delta_3}^{\Delta_4} \frac{1}{A\varepsilon} f(\frac{t}{\varepsilon}) dt$$

The first two integrals contain the origin and hence by item 2 above they evaluate to 1. Therefore, the third integral must evaluate to zero.

$$\lim_{\varepsilon \to 0} \int_{\Delta_3}^{\Delta_4} \frac{1}{A\varepsilon} f(\frac{t}{\varepsilon}) dt = 0$$

4. We can similarly show that for the interval $[-\Delta_5, -\Delta_6]$, for any $\Delta_5 > \Delta_6 > 0$ which does not contain the point $t = 0$. Then

$$\lim_{\varepsilon \to 0} \int_{-\Delta_5}^{\Delta_6} \frac{1}{A\varepsilon} f\left(\frac{t}{\varepsilon}\right) dt = 0 \quad \square$$

More complicated examples

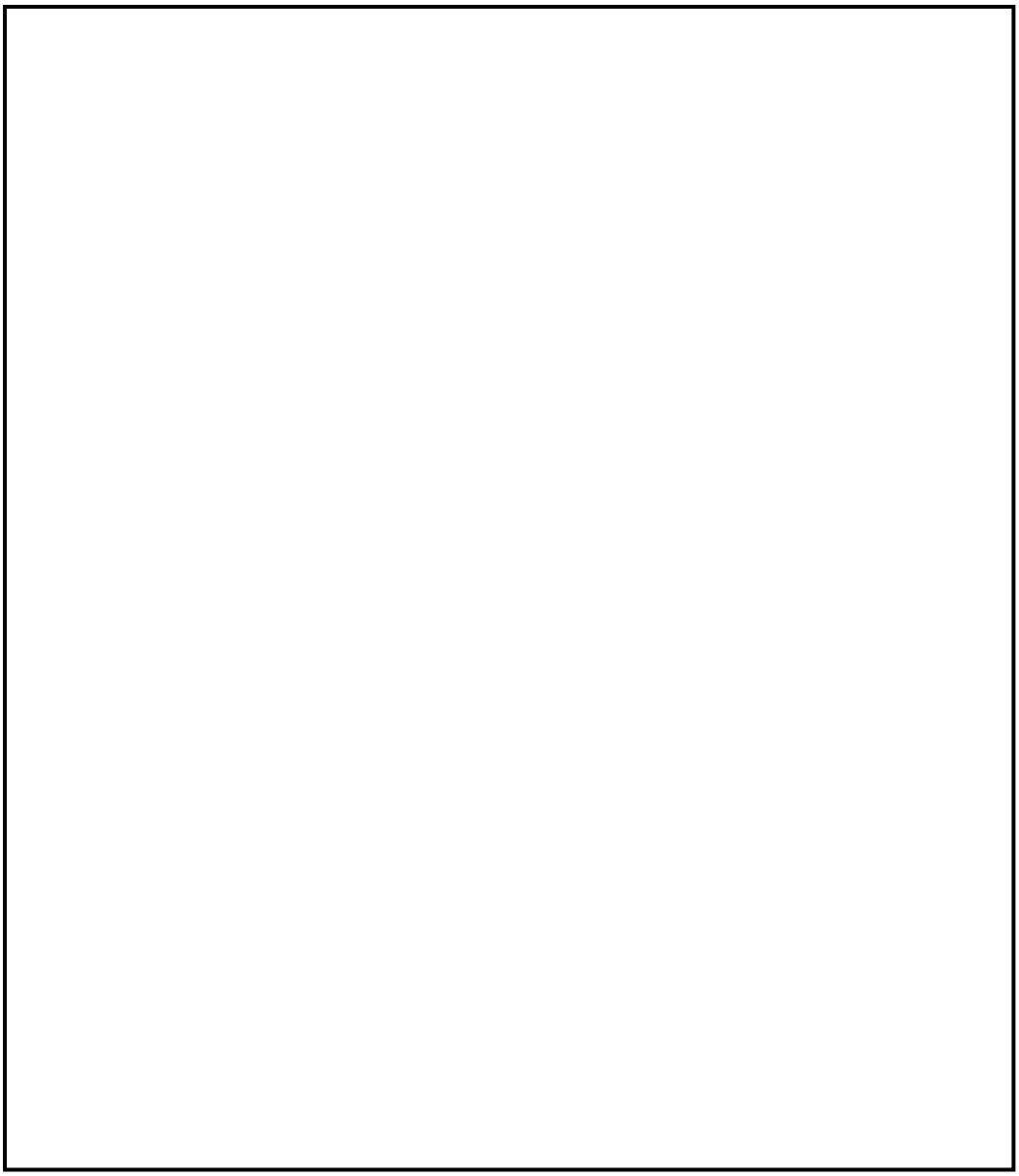
Example 1: Consider the function e^{-t^2}. The area under this function is

$$\int_{-\infty}^{\infty} e^{-t^2} dt = \sqrt{2\pi}$$

Therefore,

$$\delta(t) = \lim_{\varepsilon \to 0} \frac{1}{\sqrt{2\pi}\varepsilon} e^{-(\frac{t}{\varepsilon})^2}$$

Plot of $\chi(\,t\,)/t$



Frequency: 1/

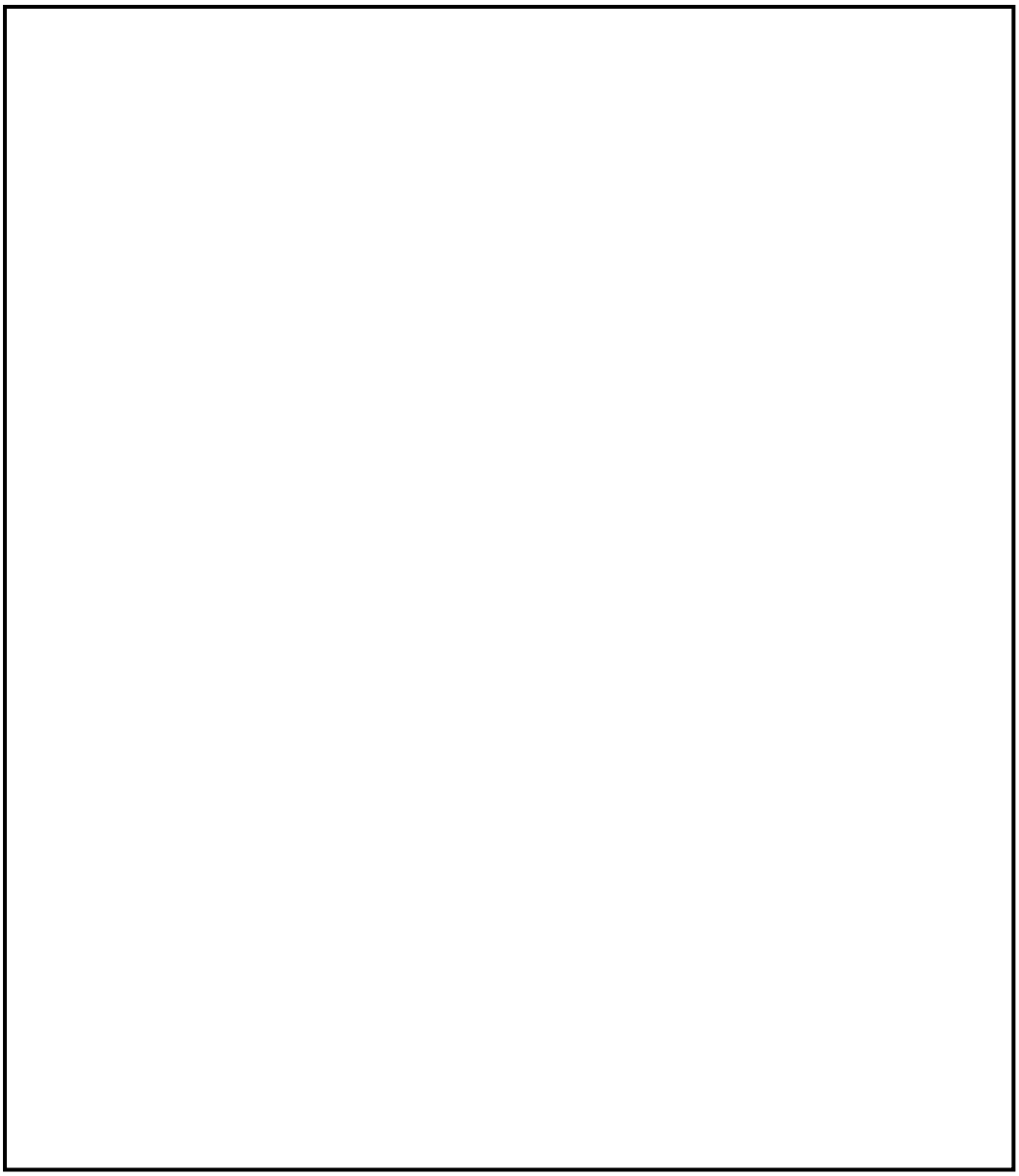
Example 2: Consider the function $e^{-(t-a)^2}$. This is the shifted version of the previous example, with the same area under it.

$$\int_{-\infty}^{\infty} e^{-(t-a)^2} dt = \sqrt{2\pi}$$

Even though this function is not peaked at $t = 0$ (it is actually peaked at $t = a$), It will still give a dirac delta peaked at t=0.

$$\delta(t) = \lim_{\varepsilon \to 0} \frac{1}{\sqrt{2\pi}\varepsilon} e^{-(\frac{t}{\varepsilon}-a)^2}$$

Plot of $\chi(\,t\,)/t$



Frequency: 1/

Example 3: Consider the function $\frac{1}{1+t^2}$. The area under this function is

$$\int_{-\infty}^{\infty} \frac{dt}{1+t^2} dt = \arctan(t)|_{-\infty}^{\infty} = \pi$$

Therefore,

$$\delta(t) = \lim_{\varepsilon \to 0} \frac{1}{\varepsilon \pi} \frac{1}{1+\left(\frac{t}{\epsilon}\right)^2}$$

Example 4: Consider the function $e^{-\alpha|t|}$, with $\alpha > 0$. The area under this function is

$$\int_{-\infty}^{\infty} e^{-\alpha|t|} dt = \frac{2}{\alpha}$$

Therefore,

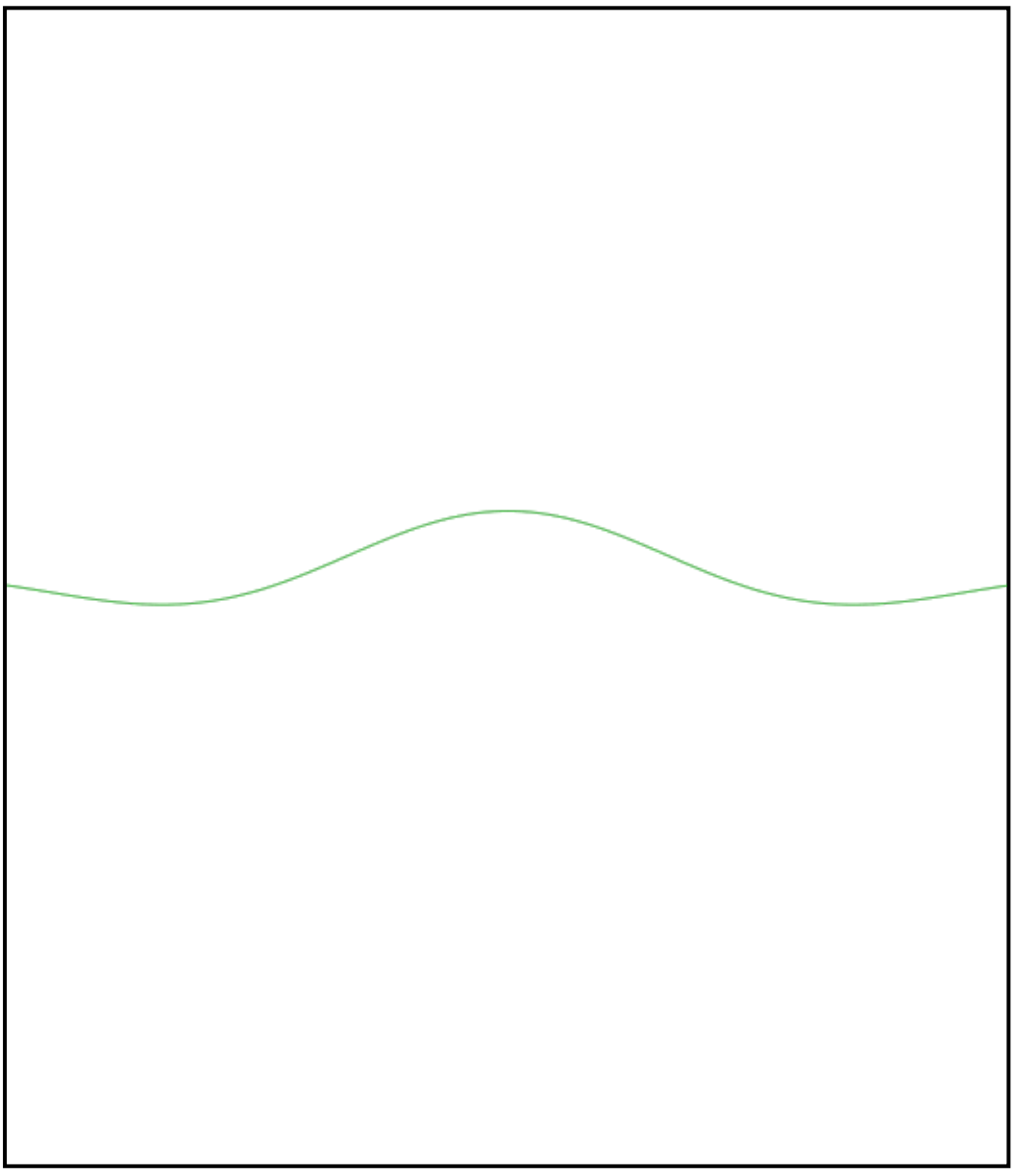
$$\delta(t) = \lim_{\varepsilon \to 0} \frac{\alpha}{2\varepsilon} e^{-\alpha \frac{|t|}{\epsilon}}$$

Example 5: Consider the sinc function. Remember that the area under it is finite and is given by the dirichlet integral:

Therefore we can costruct a dirac delta function out of it:

$$\delta(t) = \lim_{\varepsilon \to 0} \frac{1}{\pi} \frac{\sin\left(\frac{t}{\varepsilon}\right)}{t}$$

Plot of $\chi(\imath t)/t$



ε: 1/1

What will happen as $\varepsilon \to 0$? The oscillations at the two sides of the origin will become infinitely fast, and they will fill more and more the space between the envelopes. After some time, our plot will look like something like this..

Note that away from the origin $\delta(t)$ is not zero as in the previous examples.. It doesnt have a well-defined value either.. In all of the previous examples, we can say that $\delta(5) = 0$. What is $\delta(5)$ in this example? We cannot say, as the function wildly oscillates and does not converge to any definite value at $t = 5$. This is exactly the case we have discussed at the start of this chapter.

This more complicated example shows that we cannot think dirac deltas as regular functions.. A function is a rule which assigns a number to a number, say, $f(5) = 3$. Dirac deltas do not behave that way. We can only talk about them in areas, not in values. Given a dirac delta, all we can say is, all the area is concentrated on top of the origin and there is no net area elsewhere. As mentioned before, such mathematical objects are called “generalized functions” or “distributions”.

Example 6: Consider

$$\frac{1}{\pi}\frac{\cos(t)}{t}$$

This is an odd function, therefore area under it is zero, ie

$$\int_{-\infty}^{\infty} \frac{1}{\pi}\frac{\cos(t)}{t} dt = 0$$

Hence we cannot construct a dirac delta out of this function. Note that the area under the following curve

$$\frac{1}{\pi}\frac{\cos\left(\frac{t}{\varepsilon}\right)}{t}$$

is always zero, independently of ε. Hence

$$\lim_{\varepsilon \to 0} \frac{1}{\pi}\frac{\cos\left(\frac{t}{\varepsilon}\right)}{t} = 0$$

in a distributional sense. (ie, the function is not zero but its integral on every interval is zero).

Example 7: Using the last two examples above we can conclude that

$$\delta(t) = \lim_{\varepsilon \to 0} \frac{1}{i\pi t} e^{i\frac{t}{\varepsilon}}$$

Properties of Dirac Delta Function

1) Forbidden operations with dirac delta.

Dirac delta functions cannot be squared. Let us illustrate this with an example

$$\begin{aligned} \delta^2(t) &= \left(\lim_{\varepsilon \to 0} \frac{1}{2\varepsilon} \chi_{[-\varepsilon,\varepsilon]}(t) \right)^2 \\ &= \lim_{\varepsilon \to 0} \frac{1}{4\varepsilon^2} \chi_{[-\varepsilon,\varepsilon]}(t) \end{aligned}$$

As remarked before, this puts an infinite area on top of the origin. Hence it is not a function or distribution.

2) Graphing dirac deltas.

3) Translation

4) Scaling: for $a \in \mathbb{R}$

$$\delta(at) = \frac{1}{|a|} \delta(t)$$

5) Multiplying with a constant

6) Multiplying with a function

Dirac deltas are either zero of infinitely fast oscillations away from origin, depending on the function from which they are constructed. In practice we may consider both cases as zero, as they dont contain any net area under them. Therefore

$$f(t)\delta(t) = f(0)\delta(t)$$

Or, more generally

$$f(t)\delta(t - t_0) = f(t_0)\delta(t - t_0)$$

Example: $t^2 \chi_{(-4,10)}(t)\delta(t-3) = 9\delta(t-3)$

7) integration

Differentiation and integration of functions.

Theorem: Derivative of unit step function is dirac delta

$$\frac{d}{dt}U(t) = \delta(t)$$

Proof: Applying the definition of derivative

$$\frac{d}{dt}U(t) = \lim_{\varepsilon \to 0} \frac{U(t+\varepsilon) - U(t-\varepsilon)}{2\varepsilon} = \lim_{\varepsilon \to 0} \frac{\chi_{[-\varepsilon,\varepsilon]}}{2\varepsilon} = \delta(t)$$

More generally, any discontinuity of

8) Integral of Dirac delta function is unit step.

Periodic Dirac Delta Functions

Consider the functions

$$P_1(t) = \frac{\sin[(N+\frac{1}{2})t]}{\sin(\frac{t}{2})}$$
$$P_2(t) = \frac{\cos[(N+\frac{1}{2})t]}{\sin(\frac{t}{2})}$$

$P_1(t)$ is periodic with a period 2π.

$P_2(t)$ is an odd function, $P_2(t) = P_2(-t)$, independently of N. Therefore

$$\int_{-A}^{A} \frac{\cos[(N+\frac{1}{2})t]}{\sin(\frac{t}{2})} dt = 0$$

for any $A \in \mathbb{R}$.

The integral of $P_1(t)$ for a single period is independent of N

$$\frac{d}{dN}\int_{-\pi}^{\pi}\frac{\sin[(N+\frac{1}{2})t]}{\sin(\frac{t}{2})}dt = (N+\frac{1}{2})\int_{-2\pi}^{2\pi}\frac{\cos[(N+\frac{1}{2})t]}{\sin(\frac{t}{2})}dt$$
$$= (N+\frac{1}{2})\int_{-2\pi}^{2\pi}P_2(t)dt = 0$$

which follows from the oddness of $P_2(t)$.

Theorem:

$$\lim_{N\to\infty}\frac{\sin[(N+\frac{1}{2})t]}{\sin(\frac{t}{2})} = 2\pi\sum_{k=-\infty}^{\infty}\delta(t-2\pi k)$$

Proof: As P_1 (t) is periodic with period 2π, it is sufficient to prove the theorem for a single period. For the interval $-\pi < t < \pi$ the above theorem reduces to

$$\lim_{N\to\infty}\frac{\sin[(N+\frac{1}{2})t]}{\sin(\frac{t}{2})} = 2\pi\delta(t)$$

Let $f(t)$ be a smooth and bounded function for $-\pi < t < \pi$. Foe $-\pi < a < 0 < b < \pi$ consider the integral

$$\int_a^b \lim_{N\to\infty}\frac{\sin[(N+\frac{1}{2})t]}{\sin(\frac{t}{2})}f(t)dt$$

+

we can separate this integral into three parts:

$$\int_a^{-\epsilon}\lim_{N\to\infty}\frac{\sin[(N+\frac{1}{2})t]}{\sin(\frac{t}{2})}f(t)dt + \int_{-\epsilon}^{\epsilon}\lim_{N\to\infty}\frac{\sin[(N+\frac{1}{2})t]}{\sin(\frac{t}{2})}f(t)dt + \int_{\epsilon}^{b}\lim_{N\to\infty}\frac{\sin[(N+\frac{1}{2})t]}{\sin(\frac{t}{2})}f(t)dt$$

- The first and third integrals are multiplications of a smooth function $\frac{f(t)}{\cos(\frac{t}{2})}$ with an infimitely fast oscillation $\lim_{N\to\infty}\sin[(N+\frac{1}{2})t]$. By the Riemann-Lebesgue theorem, the result is zero.
- The second integral is taken in a very narrow region around zero. In this region $-f(t)\approx f(0)$

$$-\sin(\tfrac{t}{2}) \approx \tfrac{t}{2}$$

therefore

$$\int_{-\epsilon}^{\epsilon} \lim_{N\to\infty} \frac{\sin[(N+\frac{1}{2})t]}{\sin(\frac{t}{2})} f(t)dt = f(0)\int_{-\epsilon}^{\epsilon} \lim_{N\to\infty} \frac{\sin[(N+\frac{1}{2})t]}{\frac{t}{2}} dt$$

but we have derived above that

$$\lim_{N\to\infty} \frac{\sin[(N+\frac{1}{2})t]}{\frac{t}{2}} = 2\pi\delta(t)$$

therefore

$$\int_{-\epsilon}^{\epsilon} \lim_{N\to\infty} \frac{\sin[(N+\frac{1}{2})t]}{\sin(\frac{t}{2})} f(t)dt = f(0)\int_{-\epsilon}^{\epsilon} 2\pi\delta(t)dt = 2\pi f(0)$$

and

$$\int_{a}^{b} \lim_{N\to\infty} \frac{\sin[(N+\frac{1}{2})t]}{\sin(\frac{t}{2})} f(t)dt = 2\pi f(0)$$

This indicates that

$$\lim_{N\to\infty} \frac{\sin[(N+\frac{1}{2})t]}{\sin(\frac{t}{2})} = 2\pi\delta(t)$$

on a single period. When we extend this result to all the periods, we prove the theorem.

Theorem:

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